

## Multiple Choice (10 marks)

1. If a, b, and c are integers, which of the following conditions is sufficient to guarantee that the expression  $a < c \parallel a < b \ \&\& \ ! (a == c)$  evaluates to true?
  - (a)  $a < c$
  - (b)  $a < b$
  - (c)  $a > b$
  - (d)  $a == b$
2. Let  $l = [1,2,3,4]$ . What is  $\text{sum}(l)$  in Python 3?
  - (a) 4
  - (b) 1
  - (c) 10
  - (d) 8
3. Let  $l = [1,2,3,4]$ . What is  $\text{max}(l)$  in Python 3?
  - (a) 1
  - (b) 2
  - (c) 3
  - (d) 4
4. The boolean expression  $a[i] == \text{max} \parallel \!(\text{max} != a[i])$  can be simplified to
  - (a)  $a[i] == \text{max}$
  - (b)  $a[i] != \text{max}$
  - (c)  $a[i] < \text{max} \parallel a[i] > \text{max}$
  - (d) FALSE
5. In Python 3, what is  $['a', 'Chemistry', 0, 1][-3]$ ?
  - (a) a
  - (b) Chemistry
  - (c) 0
  - (d) 1

## True / False (10 marks)

*Answer True or False*

6. True or False: In Python 3, using `b[::2]` on the list `[11,13,15,17,19,21]` returns `[11,15,19]`.
7. True or False: In Python 3, the function `min([1, 2, 3, 4])` returns the second smallest value in the list, which is 2.
8. True or False: In Python 3, the function `choice(seq)` sets the integer starting value used in generating random numbers.
9. True or False: In Python 3, the expression `'a' + 'ab'` results in the string `'aab'`.
10. True or False: In Python 3, accessing the first element of the tuple `( 'abcd', 786, 2.23, 'john', 70.2 )` will output `'abcd'`.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the implications of failing to check for a zero value in the denominator when calculating the average of N numbers in an algorithm. When would this error become apparent?
12. Given the list `l = [1,2,2,3,4]`, explain how the Python 3 set function processes this list to produce its output. Discuss the significance of using sets in programming.

*End of Paper*